Syllabus
Software Engineering Department

Course Name	OBJECT-ORIENTED ANALYSIS AND DESIGN
Course No.	3502864
Lecturers Name	Vladimir Soroka vsoroka@gmail.com
Teaching Assistants	Vladimir Soroka vsoroka@gmail.com
Year	2025-2026
Weekly Hours	4 (2 + 2)
Credits	3
Prerequisite	Programming 3 - Object-Oriented Programming Software Requirements

Course Summary
Continuation of Object-Oriented Programming covering smart pointers, reference counting, software design principles, UML diagrams, and design patterns.

Learning Methods
Frontal lectures, homework, exercises, self-study, quizzes, project, final exam (face-to-face)

Learning Outcomes
Full knowledge of analysis and design of complex systems and ability to solve complex development problems.

Component	Weight
Project	40%
Assignments	10%
Quizzes	10%
Exam	40%

Topic	Notes	**Assignments**
1 – 26.3	Smart pointers	
2 – 9.4	Reference counting	
3 – 16.4	Class relationships, UML intro	HW1
4 – 23.4	Use case, sequence, communication, activity	
5 – 30.4	State, object, package, component, deployment	
6 – 7.5	Principles of software design	HW2
7 – 14.5	Singleton, Prototype	Quiz1
8 – 28.5	Abstract Factory, Builder, Composite	
9 – 4.6	Composite, Decorator	
10 – 11.6	Proxy, Observer	HW3
11 – 18.6	Template Method, Strategy, Iterator	Quiz2
12 – 25.6	Aspect-Oriented Programming	Project

Bibliography

More Effective C++, Scott Meyers, 1997

Object-Oriented Analysis and Design with Applications, Booch, 2000

Design Patterns, Gamma et al., 1995

Eclipse AspectJ, Colyer, 2005